Muhammad Asad

Data Engineer

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Professional Summary

Data Engineer with hands-on experience building cloud-native ETL/ELT pipelines using Azure Data Factory, Databricks, and PySpark. Skilled in designing Medallion Architecture lakehouses, implementing Delta Live Tables, and enforcing data governance with Unity Catalog. Strong foundation in data modeling, SQL, and end-to-end analytics solutions, with a track record of building reliable, production-ready data pipelines across multiple end-to-end projects.

Technical Skills & Core Competencies

Cloud & Data Platforms:	Azure Data Factory, Azure Data Lake Storage Gen2, Azure Databricks, Azure SQL Database, Unity Catalog, AWS (fundamentals), Microsoft Fabric
Big Data & Processing:	PySpark, Apache Spark, Delta Lake, Delta Live Tables, Autoloader, Spark Structured Streaming
ETL/ELT & Warehousing:	ETL/ELT Pipeline Design, Data Lakehouse & Warehouse Concepts, Data Quality & Testing, Query Optimization
Languages & Frameworks:	Python, SQL, DAX, Django, JavaScript, React
DevOps & Tools:	Git, CI/CD, Docker, Logic Apps, Power BI, PostgreSQL, Apache Airflow, dbt

Professional Experience

Python Intern *July 2024 – Sep 2024*
ZIGRON *Islamabad, Pakistan*

- Built and maintained Python-based ETL scripts for backend data handling, automation, and internal tooling.
- Developed and optimized internal data tools, improving pipeline efficiency, code maintainability, and reliability.
- Assisted in debugging, testing, and deploying Python-based services, following Git version control best practices.

Projects

NYC Taxi End-to-End Data Engineering — *ADF, Databricks, PySpark, Delta Lake, Power BI*

- Built a cloud ETL/ELT pipeline using Azure Data Factory for orchestration and ADLS Gen2 as the central data lake with Medallion Architecture (Bronze, Silver, Gold).
- Ingested and transformed large datasets from cloud storage using PySpark on Azure Databricks with Delta Lake for ACID transactions and time travel, optimizing workflows for performance.
- Secured Databricks-to-Data Lake access via Azure Service Principal; validated data quality across layers and visualized curated data in Power BI.

Netflix Azure End-to-End Data Engineering — *ADF, ADLS Gen2, Databricks, DLT, Unity Catalog, Power BI*

- Designed parameterized ADF pipelines to ingest data via HTTP/REST APIs into a Medallion Architecture Lakehouse.
- Used Databricks Autoloader for incremental ingestion and PySpark for schema casting and window function transformations.
- Built reliable Gold-layer pipelines with Delta Live Tables and data quality Expectations for monitoring and testing; enforced governance via Unity Catalog.

Spotify Azure End-to-End Data Engineering — *ADF, Databricks, DLT, Unity Catalog, Logic Apps, CI/CD*

- Built real-time, dynamic ADF pipelines with backfilling to move data from Azure SQL Database (data warehouse) to ADLS Gen2 Lakehouse (Bronze/Silver/Gold).
- Implemented Spark Structured Streaming with Autoloader and SCD Type 2 logic via Delta Live Tables for Gold-layer ETL pipelines.
- Used Jinja templating for metadata-driven SQL views, Logic Apps for pipeline failure alerts, and Databricks Asset Bundles for CI/CD and documentation via Git.

Walmart Data Engineering Pipeline — *PostgreSQL, dbt, Apache Airflow, Docker, Databricks SDK*

- Designed an end-to-end ELT pipeline for Walmart retail data using PostgreSQL, dbt, and Apache Airflow, structured around a Medallion Architecture (Bronze, Silver, Gold) to progressively refine raw data into business-ready analytics.
- Built dbt models to clean, standardize, and validate raw data into reusable Silver-layer business entities (customer, order, product, employee) and curated Gold-layer fact and dimensional tables for reporting.
- Orchestrated the full pipeline with containerized Airflow DAGs (Docker), integrating the Databricks SDK to trigger and manage a Databricks-based ingestion job as part of the workflow.

Flight Delay Prediction Pipeline — *Databricks, dbt, Airflow, Docker, FastAPI, Power BI*

- Built a production-style pipeline on BTS On-Time Performance data (10.5M+ flight records) through a Medallion Architecture using dbt-databricks, with a self-determining incremental ingestion notebook that detects and loads only missing months into Bronze.
- Modeled a star schema (fact/dimension) in the Gold layer with SCD2 snapshots for airport/carrier dimensions, and Dockerized Airflow DAGs chaining ingestion, dbt snapshot/run/test with automated failure alerting.
- Exposed the same Gold-layer warehouse through two consumers: a FastAPI + LLM text-to-SQL agent (Streamlit UI) for natural-language queries, and a Power BI dashboard on live DirectQuery, both always reflecting the latest pipeline run with no manual refresh.

Merit Order ETL Pipeline — *Databricks, PySpark, Delta Lake, Unity Catalog, ai_parse_document*

- Built a Bronze/Silver/Gold pipeline on Databricks tracking plant-wise fuel costs on Pakistan's power grid from NTDC's Merit Order/EMO reports; extracted PDF download links directly from the source site's compiled Angular JS bundle, since the page exposes no scrapeable HTML.
- Used Databricks' `ai_parse_document` to OCR scanned, image-based report tables into a structured 8-column schema, then applied PySpark and Delta Lake for data quality validation, type standardization, and cost-breakdown calculations in the Silver layer.
- Automated the pipeline as a chained Databricks Job (Ingestion → Transformation → Insights) and computed per-fuel-type IQR outlier detection plus plant- and fuel-level cost variation in the Gold layer, distinguishing market-wide cost shocks from plant-specific anomalies.

Certifications

- **Microsoft Certified:** Fabric Analytics Engineer Associate (DP-600)
- **Microsoft Certified:** Fabric Data Engineer Associate (DP-700)
- **IBM** Data Engineer Professional Certificate
- **Microsoft** Azure SQL Certificate
- **Google** IT Automation with Python

Education

Bachelor in Software Engineering *Sep 2021 – July 2025*
HITEC University, Taxila, Pakistan

Intermediate in Pre-Engineering *2017 – 2019*
HITEC College, Taxila, Pakistan